

CodeCanvas

ABSTRACT

Project Title: CodeCanvas – Visual Code Sharing Platform for Developers

CodeCanvas is a scalable visual code-sharing platform designed to help developers discover, share, and reuse code snippets through an intuitive and visually driven interface. Traditional code-sharing platforms often focus on complete repositories, while visual platforms showcase user interfaces without providing the underlying implementation. This creates a gap between visual inspiration and practical code reuse. CodeCanvas bridges this gap by allowing developers to upload code snippets along with screenshots or UI previews, enabling users to understand both the functionality and appearance of a solution in a single place.

The platform provides advanced search capabilities that allow developers to discover relevant code snippets using keywords, programming languages, frameworks, and tags. It also includes a fork/remix mechanism that enables users to duplicate existing code snippets, customize them, and publish improved versions while maintaining relationships with the original content. This encourages collaboration, knowledge sharing, and faster development practices within the developer community.

To support scalability and maintainability, the platform is built using a microservices architecture powered by Spring Boot and Spring Cloud. Core functionalities are divided into independent services such as User Service, Post Service, Search Service, Notification Service, and Recommendation Service. Apache Kafka is used for event-driven communication between services, ensuring efficient asynchronous processing of tasks such as search indexing, notifications, and activity updates. PostgreSQL serves as the primary database for structured application data, while Redis is used for caching frequently accessed information. Elasticsearch enables fast and efficient full-text search capabilities across code snippets and tags. The frontend is developed using React.js, providing a responsive and modern user experience with real-time interactions supported through WebSockets.

The project demonstrates modern software engineering principles including microservices architecture, event-driven systems, distributed communication, caching, search optimization, and real-time application development. CodeCanvas aims to improve developer productivity by creating a centralized platform that combines visual learning, code discovery, collaboration, and reusable software components in a scalable and industry-relevant solution.